



Royal University of Phnom Penh

Faculty of Engineering

Data Science and Engineering

Course: Data Analytics

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RFM Analysis of Customer Transactions:

**A Comprehensive Study of Customer Segmentation
and Behavior**

Class: DSE Cohort 2

Team Members: Phork Norak and Nhor Povketya

Abstract

This report presents a detailed study of customer segmentation using the Recency, Frequency, and Monetary (RFM) model. We analyzed a dataset of 4,906 customer transactions from 2001 to 2006, starting with exploratory data analysis (EDA) to understand the data's patterns and clean it for better results. Using the RFM method, we grouped customers based on their buying habits and compared two different segmentation criteria to find the best way to categorize them. This process helped us sort customers into meaningful groups, providing useful insights for creating better marketing plans and improving customer relationships. Our study highlights customer behaviors, identifies key groups, and offers clear suggestions to target them effectively, aiming to boost sales and loyalty.

1. Introduction

1.1 Background of RFM Analysis

RFM (recency, frequency, monetary) analysis is a behavior based technique used to segment customers by examining their transaction history such as

- how recently a customer has purchased (recency)
- how often they purchase (frequency)
- how much the customer spends (monetary)

It is based on the marketing axiom that 80% of your business comes from 20% of your customers. RFM helps to identify customers who are more likely to respond to promotions by segmenting them into various categories.

1.2 Importance of Customer Segmentation

In today's tough market, knowing what customers want and how they behave is key to growing a business. Customer segmentation means dividing a mixed group of customers into smaller, similar groups. This helps businesses in many ways:

- Create marketing plans that work better and get more sales
- Use resources wisely to get the best results
- Offer personalized experiences to customers
- Find the most valuable customers to keep them happy
- Spot new groups that can help the business grow

Without good segmentation, companies might use general plans that don't meet the needs of different customers. This can waste resources and make it harder to keep customers interested.

1.3 Research Objectives

This study aims to achieve the following goals using the dataset (rfm_data_orders.rda.csv):

- Use the RFM method to group customers based on their buying patterns
- Identify different customer groups and understand how they behave
- Look at how customers are spread across these groups
- Suggest marketing plans for each customer group
- Study changes in customer behavior from 2001 to 2006
- Give useful ideas for improving customer relationships

2. Methodology

2.1 Data Collection

- **Timeframe:** March 6, 2004, to August 14, 2006.
- **Sample Size:** 4,906 records, 995 unique customers.
- **Data Structure:** Customer ID, order date, revenue, names, and email information.

	Unnamed: 0	customer_id	order_date	revenue	first_name	last_name
0	0	Brion Stark	2004-12-20	32	Brion	Stark
1	1	Ethyl Botsford	2005-05-02	36	Ethyl	Botsford
2	2	Hosteen Jacobi	2004-03-06	116	Hosteen	Jacobi
3	3	Edw Frami	2006-03-15	99	Edw	Frami
4	4	Josef Lemke	2006-08-14	76	Josef	Lemke

2.2 Data Preprocessing and Feature Extraction

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4906 entries, 0 to 4905
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      4906 non-null  int64
1   customer_id     4906 non-null  object
2   order_date      4906 non-null  object
3   revenue         4906 non-null  int64
4   first_name      4906 non-null  object
5   last_name       4906 non-null  object
6   email           4906 non-null  object
dtypes: int64(2), object(5)
memory usage: 268.4+ KB
```

1. **Data Type Conversion:**
 - o Converted order_date from string to datetime format
2. **Feature Selection:**
 - o Select column: customer_id, order_date, revenue
3. **Unique Year:**
 - o 2001, 2004, 2005, 2006
4. **Check Unique customer_id in 2001:**
 - o There's only one
5. **Select important years:**
 - o Choose only 2004, 2005, 2006

```
<class 'pandas.core.frame.DataFrame'>
Index: 4905 entries, 0 to 4905
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   customer_id     4905 non-null  object
1   order_date      4905 non-null  datetime64[ns]
2   revenue         4905 non-null  int64
dtypes: datetime64[ns](1), int64(1), object(1)
memory usage: 153.3+ KB
```

2.3 Building the RFM Model

1. **Recency Calculation:**
 - o Used the day after the latest order date as analysis_date (2006-12-31)
 - o Calculated days since the last purchase for each customer
2. **Frequency Calculation:**
 - o Counted the number of orders per customer
3. **Monetary Calculation:**
 - o Summed total revenue per customer

	customer_id	Recency	Frequency	Monetary
0	Abbey O'Reilly	205	6	472
1	Add Senger	140	3	340
2	Aden Lesch	194	4	405
3	Aden Murphy	98	7	596
4	Admiral Senger	132	5	448

2.4 Assigning Scores

Assign RFM scores using quintiles (5 groups, scores 1–5)

The process assigns RFM scores (1 to 5) to customers based on their Recency, Frequency, and Monetary values by dividing them into 5 equal groups (quintiles, ~20% each).

- **Recency Score:** Splits days since last purchase into 5 groups. Most recent purchases get 5, least recent get 1.
- **Frequency Score:** Ranks customers by number of purchases, splits ranks into 5 groups. Fewest purchases get 1, most get 5.
- **Monetary Score:** Ranks customers by total spending, and splits ranks into 5 groups. The lowest spending gets 1, the highest gets 5.

	customer_id	Recency	Frequency	Monetary	R_Score	F_Score	M_Score
0	Abbey O'Reilly	205	6	472	3	4	3
1	Add Senger	140	3	340	4	1	2
2	Aden Lesch	194	4	405	3	2	3
3	Aden Murphy	98	7	596	5	4	4
4	Admiral Senger	132	5	448	4	3	3

Combined RFM Score:

- Calculated using formula: $(\text{Recency Score} \times 100) + (\text{Frequency Score} \times 10) + \text{Monetary Score}$
- Example: A customer with R=5, F=5, M=5 would have RFM score of 555

	customer_id	Recency	Frequency	Monetary	R_Score	F_Score	M_Score
0	Abbey O'Reilly	205	6	472	3	4	3
1	Add Senger	140	3	340	4	1	2
2	Aden Lesch	194	4	405	3	2	3
3	Aden Murphy	98	7	596	5	4	4
4	Admiral Senger	132	5	448	4	3	3

3. Segmentation and Virtualization

3.1 Segment Criteria

Let us classify our customers based on the individual recency, frequency and monetary scores.

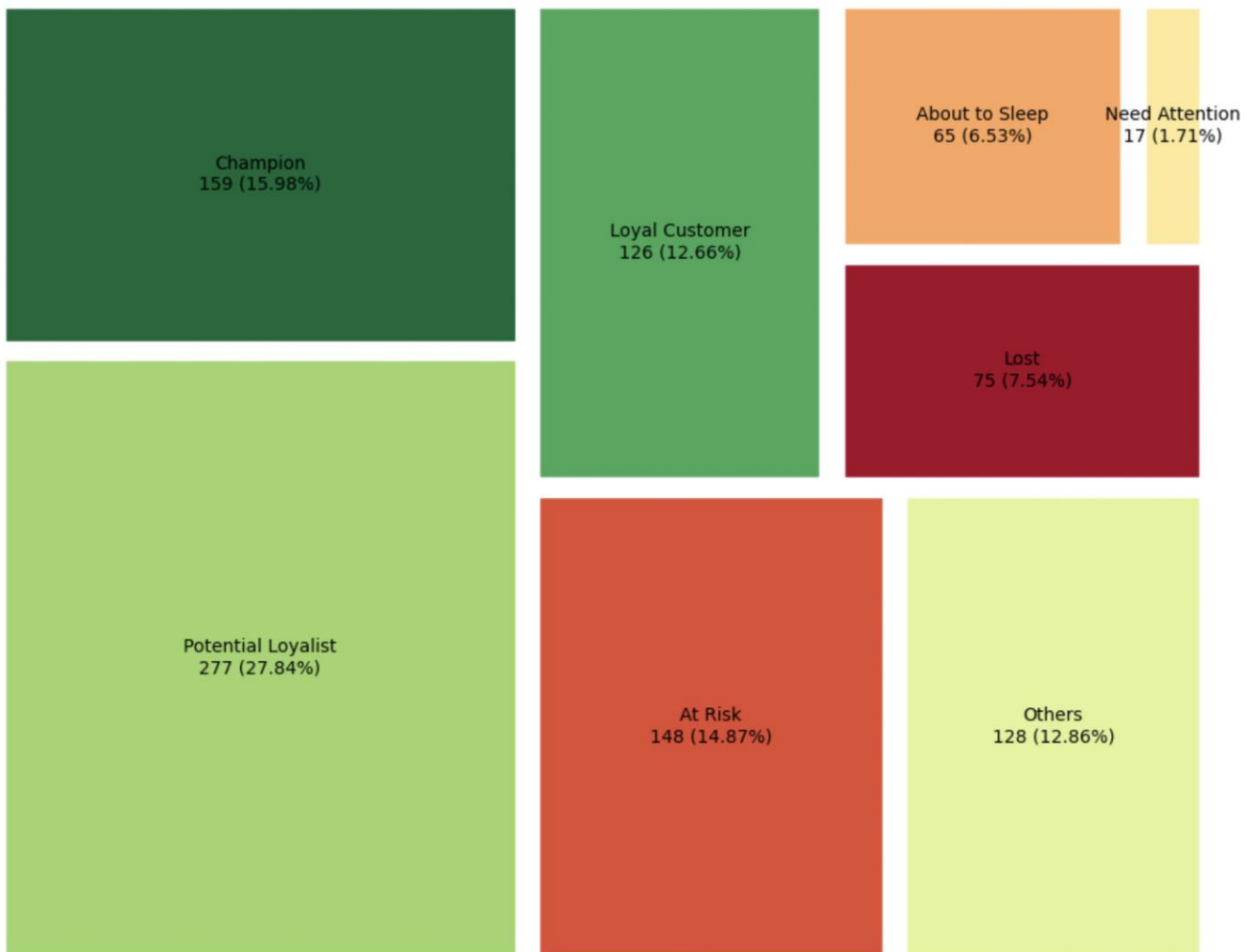
<https://blog.rsquaredacademy.com/customer-segmentation-using-rfm-analysis/>

Segment	Description
Champions	Bought recently, buy often and spend the most
Loyal Customers	Spend good money. Responsive to promotions
Potential Loyalist	Recent customers, spent good amount, bought more than
New Customers	Bought more recently, but not often
Promising	Recent shoppers, but haven't spent much
Need Attention	Above average recency, frequency & monetary values
About To Sleep	Below average recency, frequency & monetary values
At Risk	Spent big money, purchased often but long time ago
Can't Lose Them	Made big purchases and often, but long time ago
Hibernating	Low spenders, low frequency, purchased long time ago
Lost	Lowest recency, frequency & monetary scores

Based on the defined criteria, customers were distributed across the following segments:

1. **Potential Loyalists:** 277 customers (28.8%)
2. **Champions:** 159 customers (16.5%)
3. **At Risk:** 148 customers (15.4%)
4. **Others: 128 customers (13.3%)**
5. **Loyal Customers:** 126 customers (13.1%)
6. **Lost:** 75 customers (7.8%)
7. **About to Sleep:** 65 customers (6.8%)
8. **Need Attention:** 17 customers (1.8%)

Customer Segments Distribution



Others: 128 customers (12.86%)? That's a lot! Is it because of dataset or segment criteria?

Let's try different segment criteria:

<https://rfm.rsquaredacademy.com/articles/rfm-customer-level-data.html#customers-by-orders>

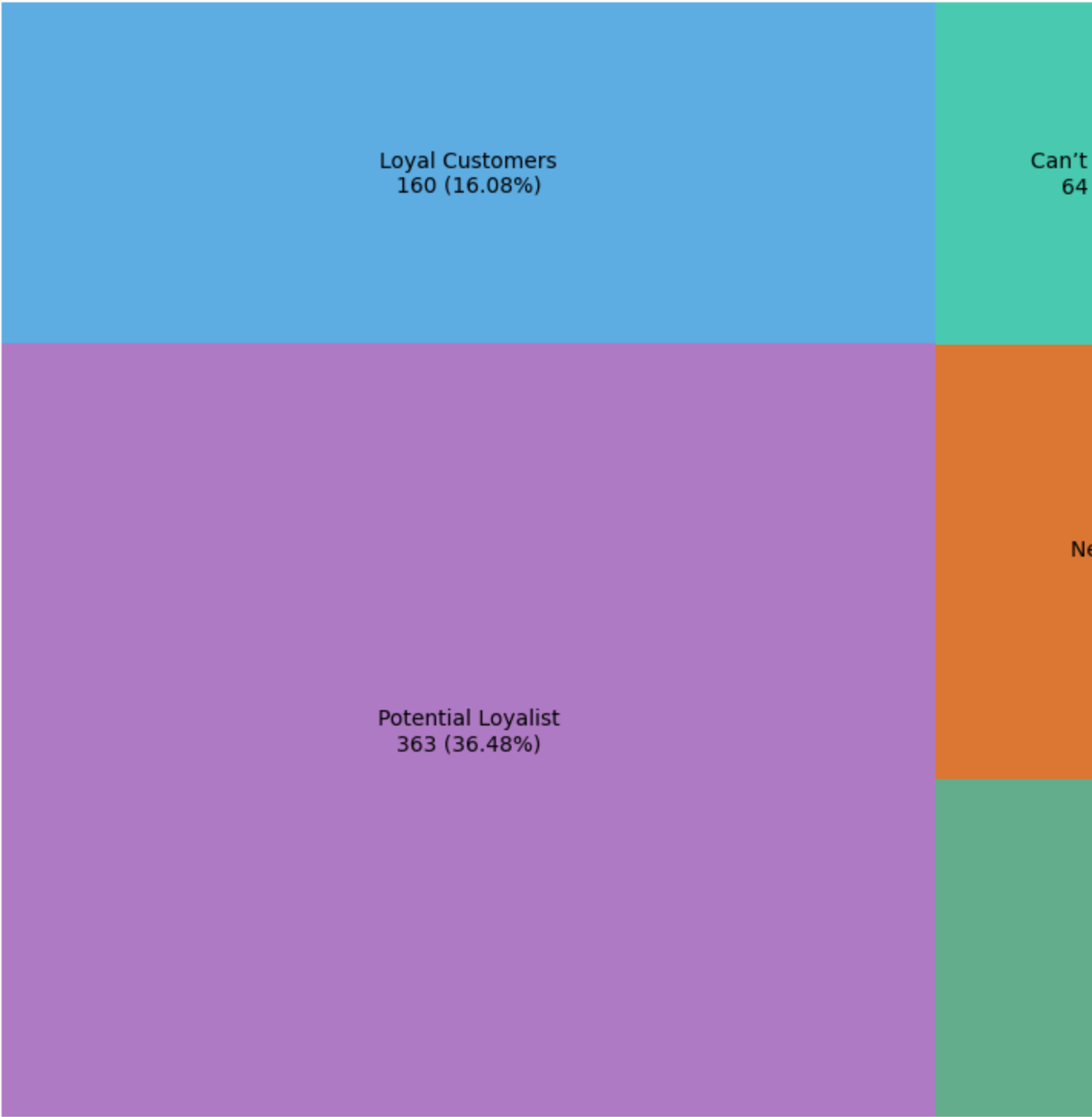
Segment	Description	R	F	M
Champions	Bought recently, buy often and spend the most	5	5	5
Potential Loyalist	Recent customers, spent good amount, bought more than once	3 - 5	3 - 5	2 - 5
Loyal Customers	Spend good money. Responsive to promotions	2 - 4	2 - 4	2 - 4
Promising	Recent shoppers, but haven't spent much	3 - 4	1 - 3	3 - 5
New Customers	Bought more recently, but not often	4 - 5	1 - 3	1 - 5
Can't Lose Them	Made big purchases and often, but long time ago	1 - 2	3 - 4	4 - 5
At Risk	Spent big money, purchased often but long time ago	1 - 2	2 - 5	4 - 5
Need Attention	Above average recency, frequency & monetary values	1 - 3	3 - 5	3 - 5
About To Sleep	Below average recency, frequency & monetary values	2 - 3	1 - 3	1 - 4
Lost	Bought a long time ago, average amount spent	1 - 1	1 - 5	1 - 5

Based on the defined criteria, customers were distributed across the following segments:

Total number of customers: 995

1. **Potential Loyalist:** 363 customers (36.48%)
2. **Loyal Customers:** 160 customers (16.08%)
3. **Lost:** 143 customers (14.37%)
4. **New Customers:** 96 customers (9.65%)
5. **About To Sleep:** 88 customers (8.84%)
6. **Can't Lose Them:** 64 customers (6.43%)
7. **Champions:** 40 customers (4.02%)
8. **At Risk:** 20 customers (2.01%)
9. **Need Attention:** 17 customers (1.71%)
10. **Promising:** 3 customers (0.30%)
11. **Other:** 1 customers (0.10%)

Customer Segments Distribu

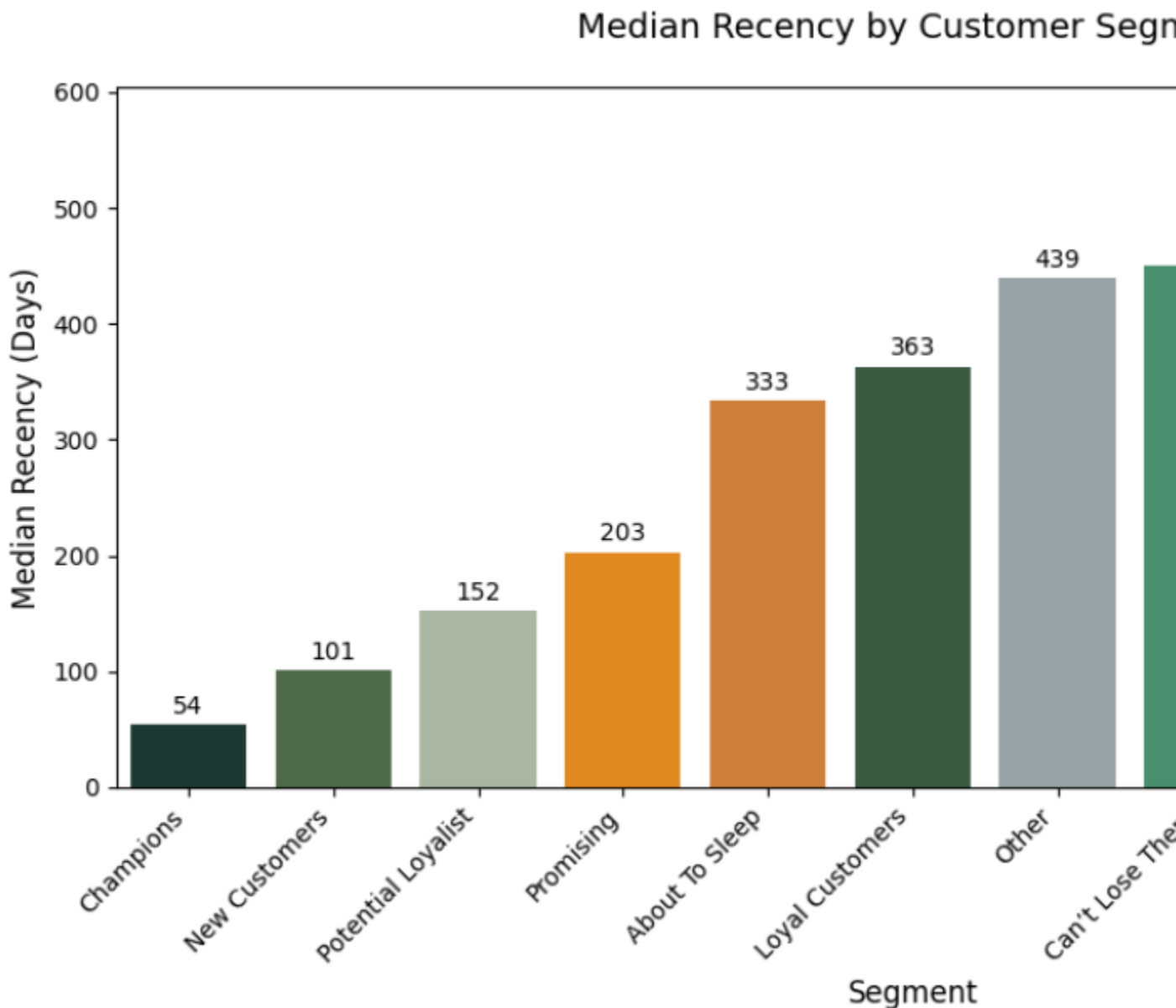


Other: 1 customer (0.10%) now it's much better. Looks like this segment criteria is more fit with our dataset.

Insights:

- The biggest group, Potential Loyalists, makes up 36.48% (363 customers). These are recent buyers who shop often and spend a good amount, so they could become loyal with the right efforts.
- On the other hand, 14.37% (143 customers) are Lost, meaning they haven't bought in a long time and need actions to bring them back.
- Smaller groups like Champions (4.02%) and Loyal Customers (16.08%) are very valuable because they spend a lot, so keeping them happy is important.

3.2 Median Recency Analysis by Segment



1. **Champions (40 customers, 4.02%) are the most active:** They bought just 54 days ago, so they shop often and are very valuable.

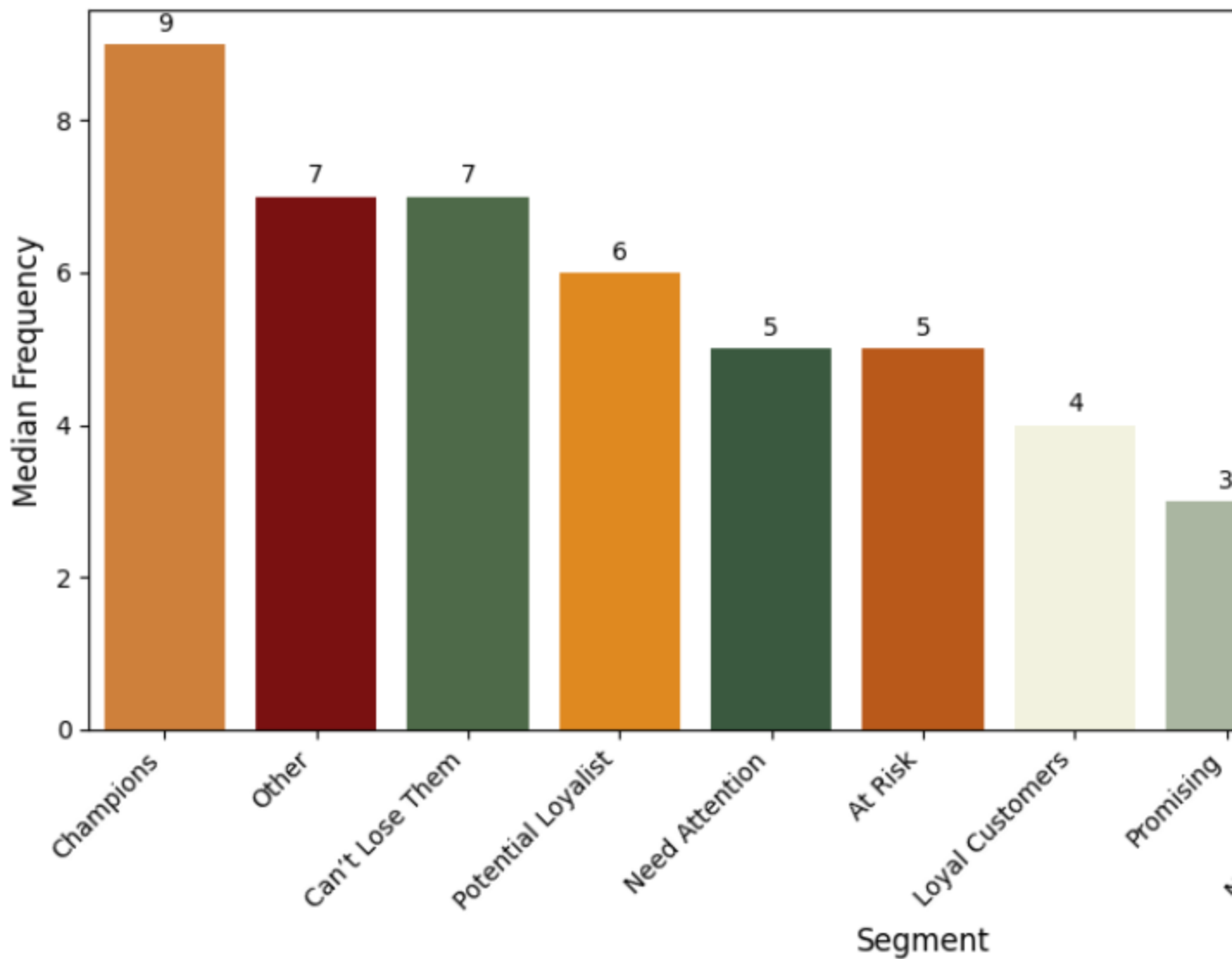
2. **New Customers (96 customers, 9.65%) are also active:** They last bought 101 days ago, showing they're new and still interested in shopping.
3. **Potential Loyalist (363 customers, 36.48%) are quite active:** They're the biggest group and haven't shopped in 152 days, so they're still buying regularly and could become loyal.
4. **Promising (3 customers, 0.30%) can do better:** They last bought 203 days ago, but there are only a few of them, so they might need a small push to shop more.
5. **About To Sleep (88 customers, 8.84%) are slowing down:** They haven't bought in 333 days, so they might stop shopping soon if we don't act.
6. **Loyal Customers (160 customers, 16.08%) are less active:** They last bought 363 days ago, so we should remind this big group to come back and keep them loyal.
7. **Other (1 customer, 0.10%) is not very active:** This single customer hasn't shopped in 439 days, so they need some attention to start buying again.
8. **Can't Lose Them (64 customers, 6.43%) are at risk:** They last bought 451 days ago, so we need to act fast to keep this group shopping.
9. **Need Attention (17 customers, 1.71%) are slipping away:** They haven't shopped in 512 days, so we should try to bring this small group back.
10. **At Risk (20 customers, 2.01%) are almost gone:** They last bought 520 days ago, so we need to work hard to get this small group shopping again.
11. **Lost (143 customers, 14.37%) are inactive:** They haven't bought in 576 days, and since it's a big group, we should try to win them back with special efforts.

Summary

Potential Loyalists are the largest and fairly active group, while Champions are small but very active. On the other hand, Lost and At Risk customers are large but haven't shopped in a long time, so they need the most effort to bring them back. Smaller groups like Promising and Need Attention also need some focus to keep them engaged.

3.3 Median Frequency Analysis by Segment

Median Frequency by Customer Segment



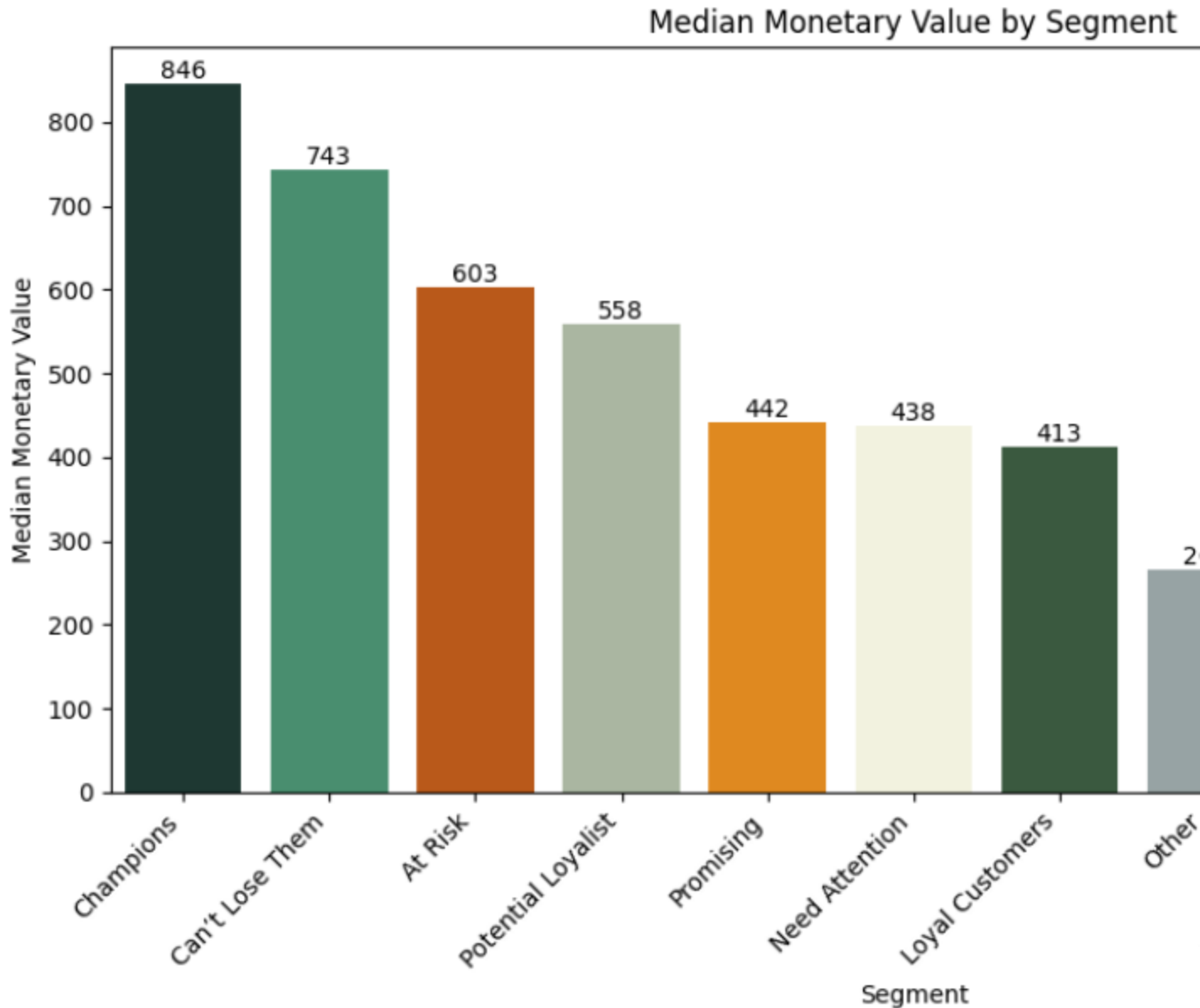
1. **Champions (40 customers, 4.02%) shop the most:** They buy 9 times on average, so this small group is very active and important.
2. **Other (1 customer, 0.10%) shops a lot:** This single customer buys 7 times on average, which is great for such a small group.
3. **Can't Lose Them (64 customers, 6.43%) shop often:** They buy 7 times on average, so we need to keep this group happy to stop them from leaving.
4. **Potential Loyalists (363 customers, 36.48%) shop quite often:** They buy 6 times on average, and since they're the biggest group, they're a great chance to make loyal customers.
5. **Need Attention (17 customers, 1.71%) shop sometimes:** They buy 5 times on average, but this small group needs more effort to keep them coming back.

6. **At Risk (20 customers, 2.01%) shop sometimes:** They also buy 5 times on average, but they might stop soon, so we should act fast.
7. **Loyal Customers (160 customers, 16.08%) shop a bit less:** They buy 4 times on average, so we should encourage this big group to shop more often.
8. **Promising (3 customers, 0.30%) don't shop much:** They buy 3 times on average, so this tiny group needs a little push to buy more.
9. **New Customers (96 customers, 9.65%) don't shop much:** They buy 3 times on average, so we should help these new shoppers buy more often.
10. **Lost (143 customers, 14.37%) don't shop much:** They buy 3 times on average, and since it's a big group, we should try to get them back.
11. **About To Sleep (88 customers, 8.84%) don't shop much:** They buy 3 times on average, so we need to wake them up before they stop shopping.

Summary

Champions and Can't Lose Them shop the most often, which is great, but they're small groups. Potential Loyalists are the biggest group and shop quite often, so they're a good chance to build loyalty. Lost and About To Sleep groups are big but don't shop much, so we need to work on bringing them back. Smaller groups like Promising and New Customers also need help to shop more.

3.4 Median Monetary Value Analysis by Segment



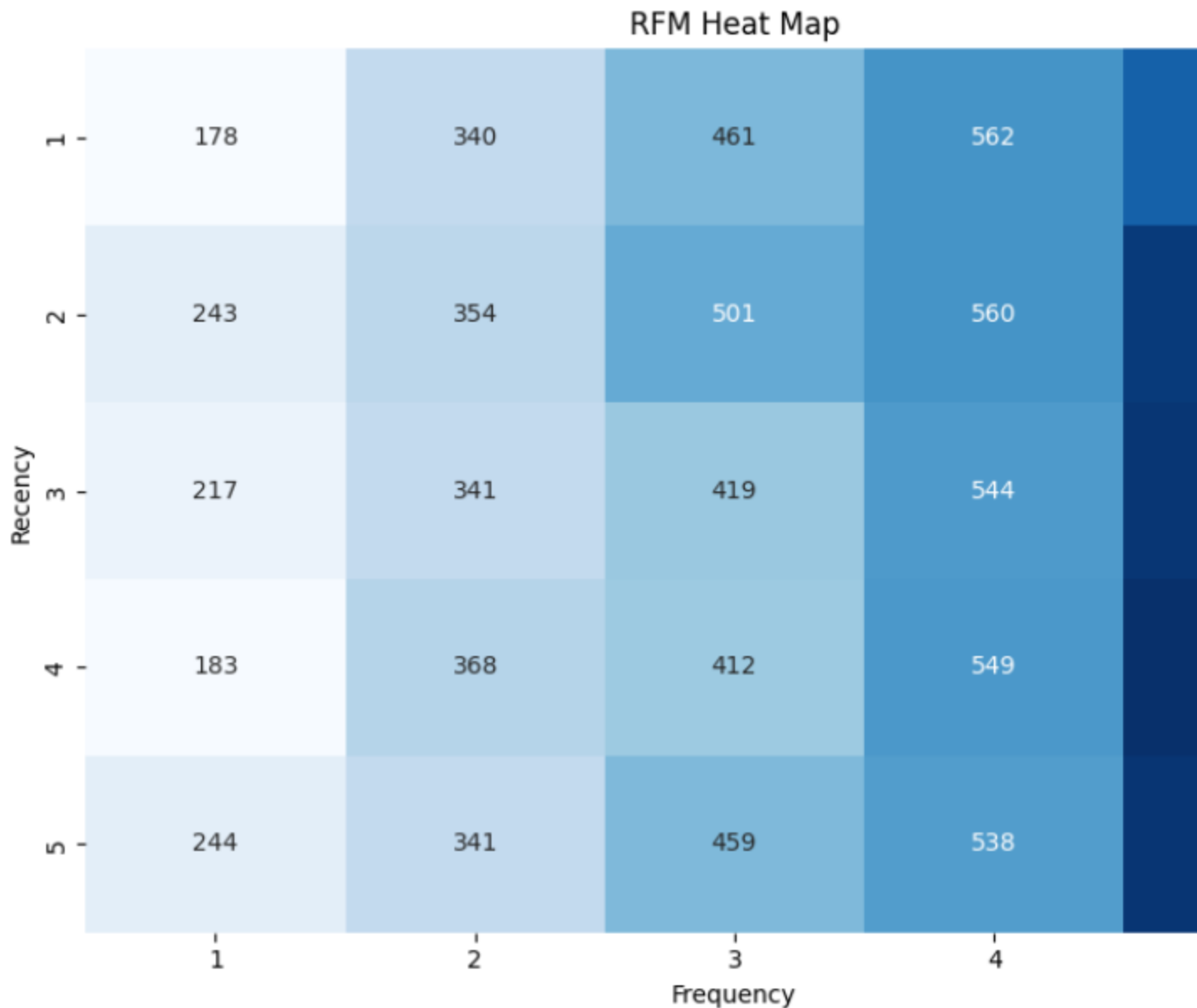
1. **Champions (40 customers, 4.02%) spend the most:** They spend 846 on average per purchase, so this small group is very valuable.
2. **Can't Lose Them (64 customers, 6.43%) spend a lot:** They spend 743 on average, so we need to keep this group happy to stop them from leaving.
3. **At Risk (20 customers, 2.01%) spend a good amount:** They spend 603 on average, but they might stop soon, so we should act fast.
4. **Potential Loyalist (363 customers, 36.48%) spend well:** They spend 558 on average, and since they're the biggest group, they bring in a lot of money.
5. **Promising (3 customers, 0.30%) spend okay:** They spend 442 on average, so this tiny group has the potential to spend more.
6. **Need Attention (17 customers, 1.71%) spend okay:** They spend 438 on average, but this small group needs more effort to keep them spending.

7. **Loyal Customers (160 customers, 16.08%) spend a fair amount:** They spend 413 on average, so we should encourage this big group to spend more.
8. **Other (1 customer, 0.10%) spends less:** This single customer spends 265 on average, which is not a lot.
9. **New Customers (96 customers, 9.65%) spend less:** They spend 241 on average, so we should help these new shoppers spend more.
10. **Lost (143 customers, 14.37%) spend the least:** They spend 222 on average, and since it's a big group, we should try to get them to spend more.
11. **About To Sleep (88 customers, 8.84%) spend the least:** They spend 215 on average, so we need to wake them up to get them to spend more.

Summary

Champions and Can't Lose Them spend the most per purchase, which is great, but they're small groups. Potential Loyalists are the biggest group and spend a good amount, so they're a great chance to make more money. Lost and About To Sleep groups are big but spend the least, so we need to work on getting them to spend more. Smaller groups like Promising and New Customers also need help to spend more.

3.5 RFM Heatmap



- **Best spenders (Recency 4, Frequency 5):** They bought a while ago but shop 5 times, spending 800 on average. These are big spenders!
- **Recent big spenders (Recency 1, Frequency 5):** They bought recently and shop 5 times, spending 681. They're great customers!
- **New but low spenders (Recency 1, Frequency 1):** They bought recently but only once, spending 178. They might spend more if they buy more often.
- **Rare and old buyers (Recency 5, Frequency 1):** They haven't bought in a long time and only shop once, spending 244. They need more encouragement.
- **Frequent but older buyers (Recency 5, Frequency 5):** They haven't bought recently but shop 5 times, spending 788. Try to get them back!
- **Middle shoppers (Recency 3, Frequency 3):** They last bought a while ago, shop 3 times, and spend 419. They could spend more with some effort.

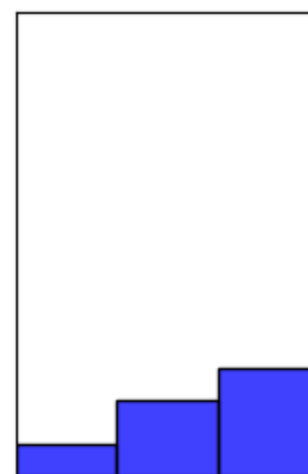
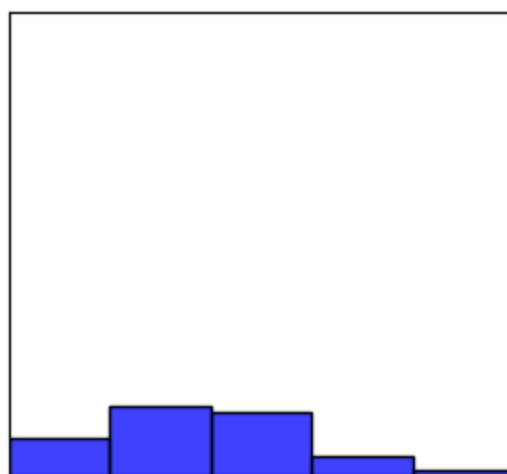
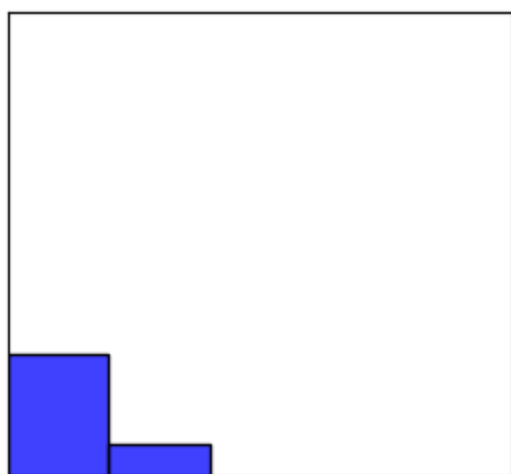
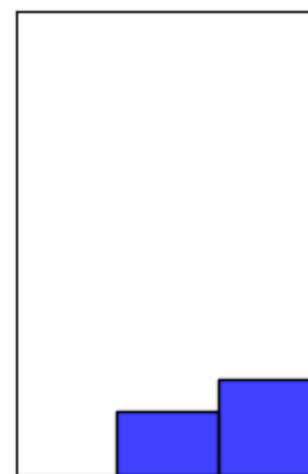
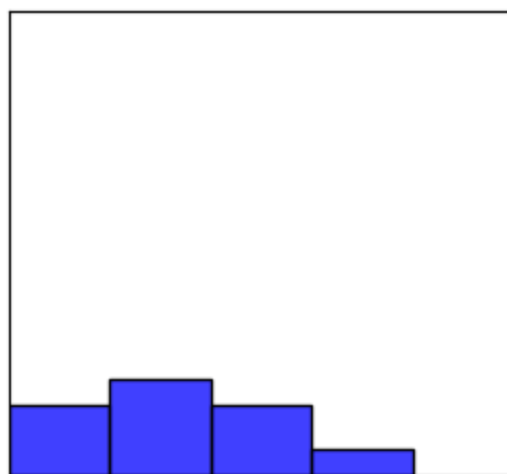
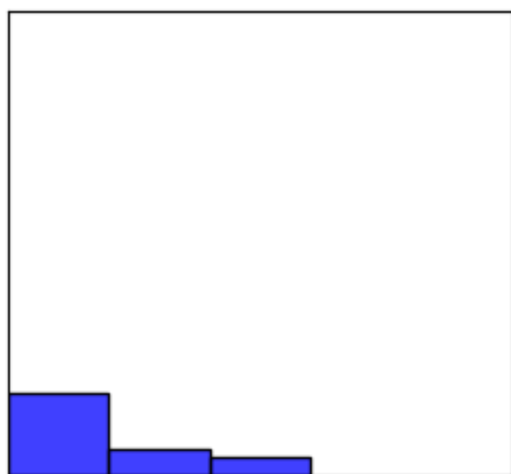
3.6 Monetary Score Distribution

Frequency

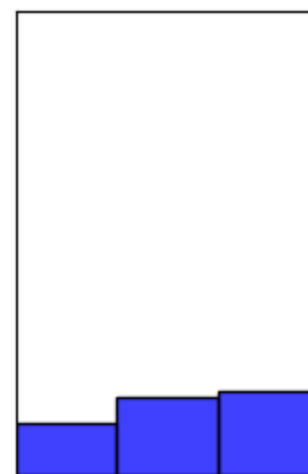
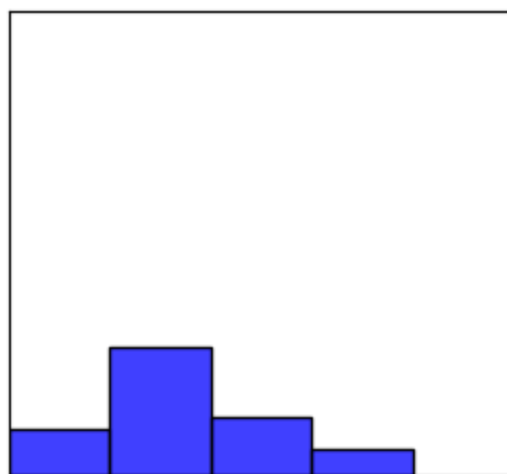
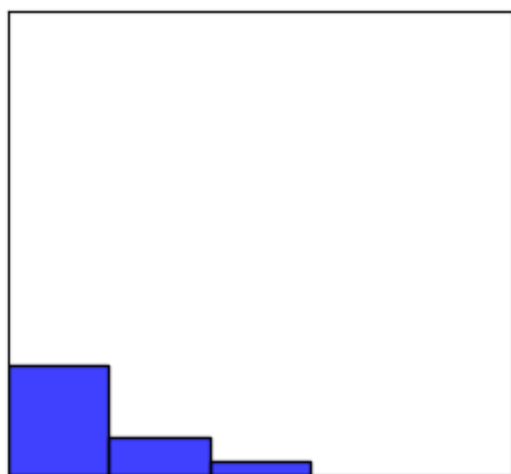
1

2

3

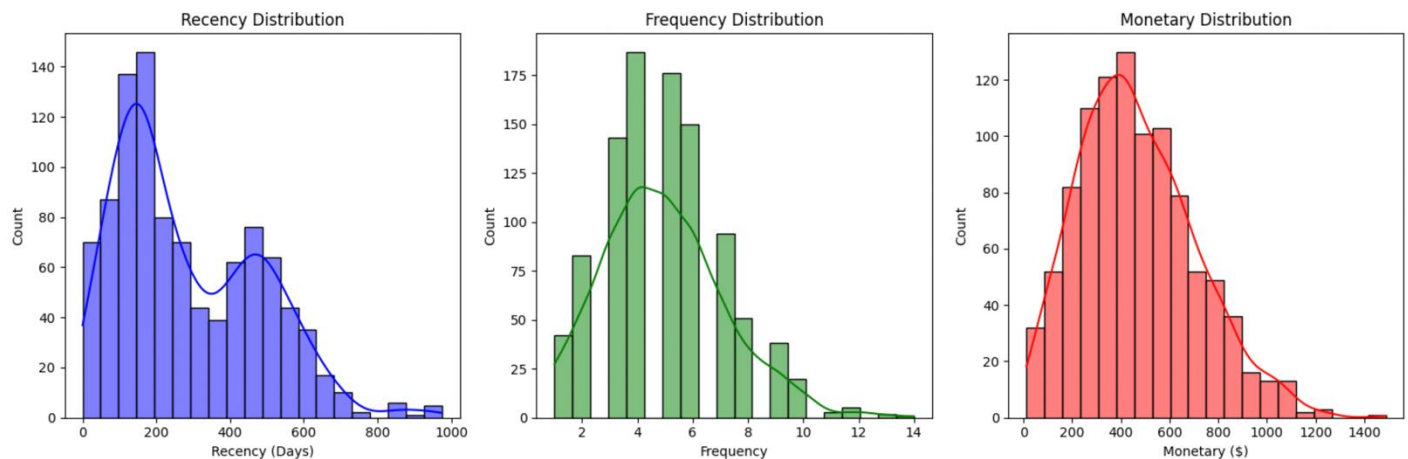


Recency Score



- **Best customers (Recency 1, Frequency 5):** Many have high Monetary scores (4–5), meaning they buy recently, often, and spend a lot.
- **New but rare buyers (Recency 1, Frequency 1):** Most have low Monetary scores (1–2), so they buy recently but not often and spend little.
- **Frequent but old buyers (Recency 5, Frequency 5):** Many have high Monetary scores (4–5), showing they used to buy often and spend a lot, but not recently.
- **Inactive and rare buyers (Recency 5, Frequency 1):** Most have low Monetary scores (1–2), meaning they don't buy often, haven't bought recently, and spend little.
- **Middle group (Recency 3, Frequency 3):** Many have medium Monetary scores (2–3), so they buy sometimes, not too recently, and spend an average amount.
- **Recent and frequent (Recency 1, Frequency 4):** Many have high Monetary scores (3–5), meaning they buy recently, quite often, and spend a good amount.

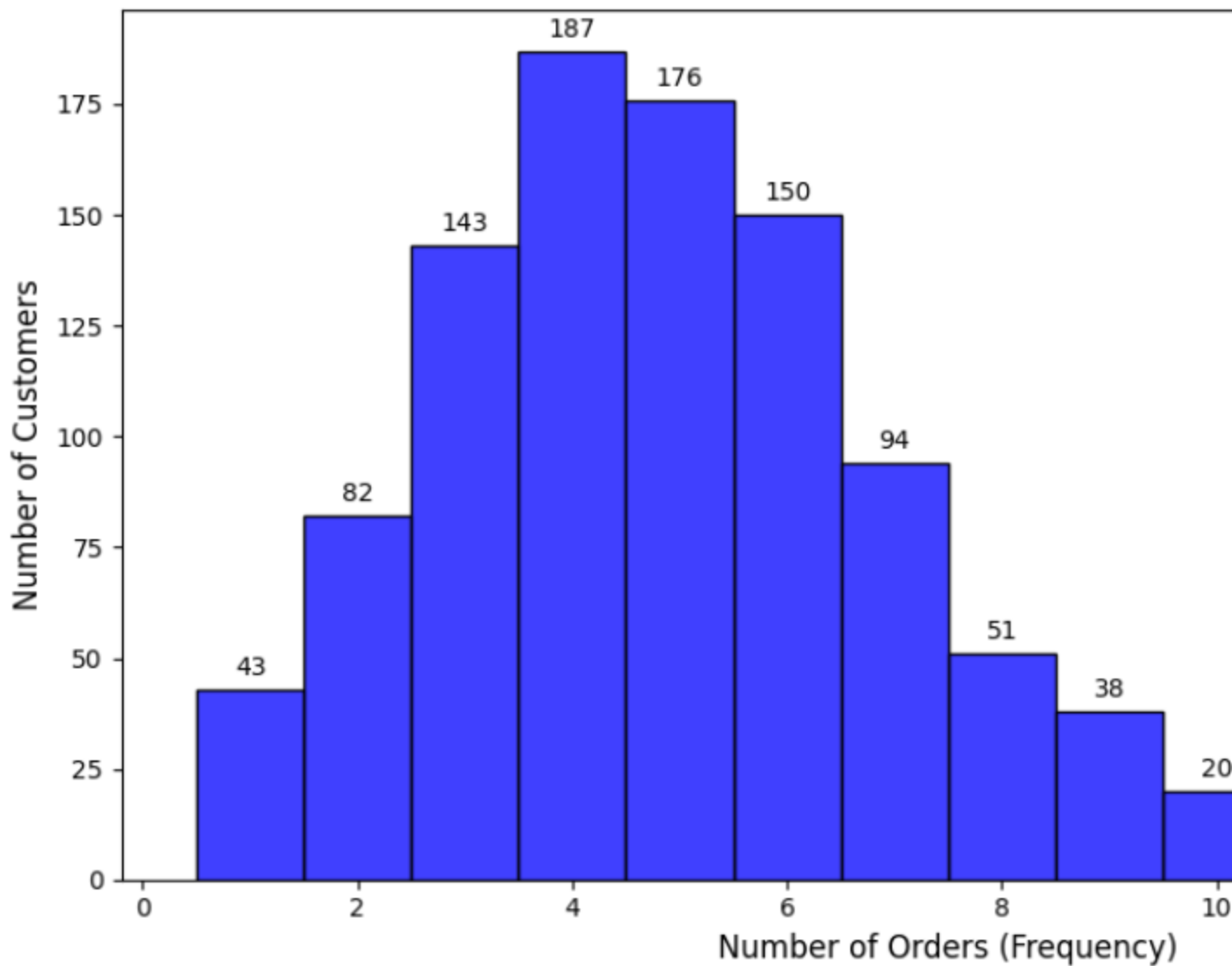
3.7 RFM Metric Distributions



- **Recency (Days):** Many customers (over 175) bought recently (0–100 days), but some haven't bought in a long time (over 800 days).
- **Frequency (How Often):** Most customers buy 2–5 times (over 150 customers), but a few buy a lot, like 10–14 times.
- **Monetary (Spending):** Many customers spend \$0–\$200 (over 100 customers), but some spend a lot, up to \$1400.
- **Recent Buyers:** A big group bought in the last 100 days, which is great for business.
- **Low Frequency:** Not many customers buy more than 5 times, so we need to encourage more frequent purchases.
- **Low Spenders:** Most customers spend less than \$200, so we should try to get them to spend more.

3.8 Purchase Frequency Analysis

Order Distribution

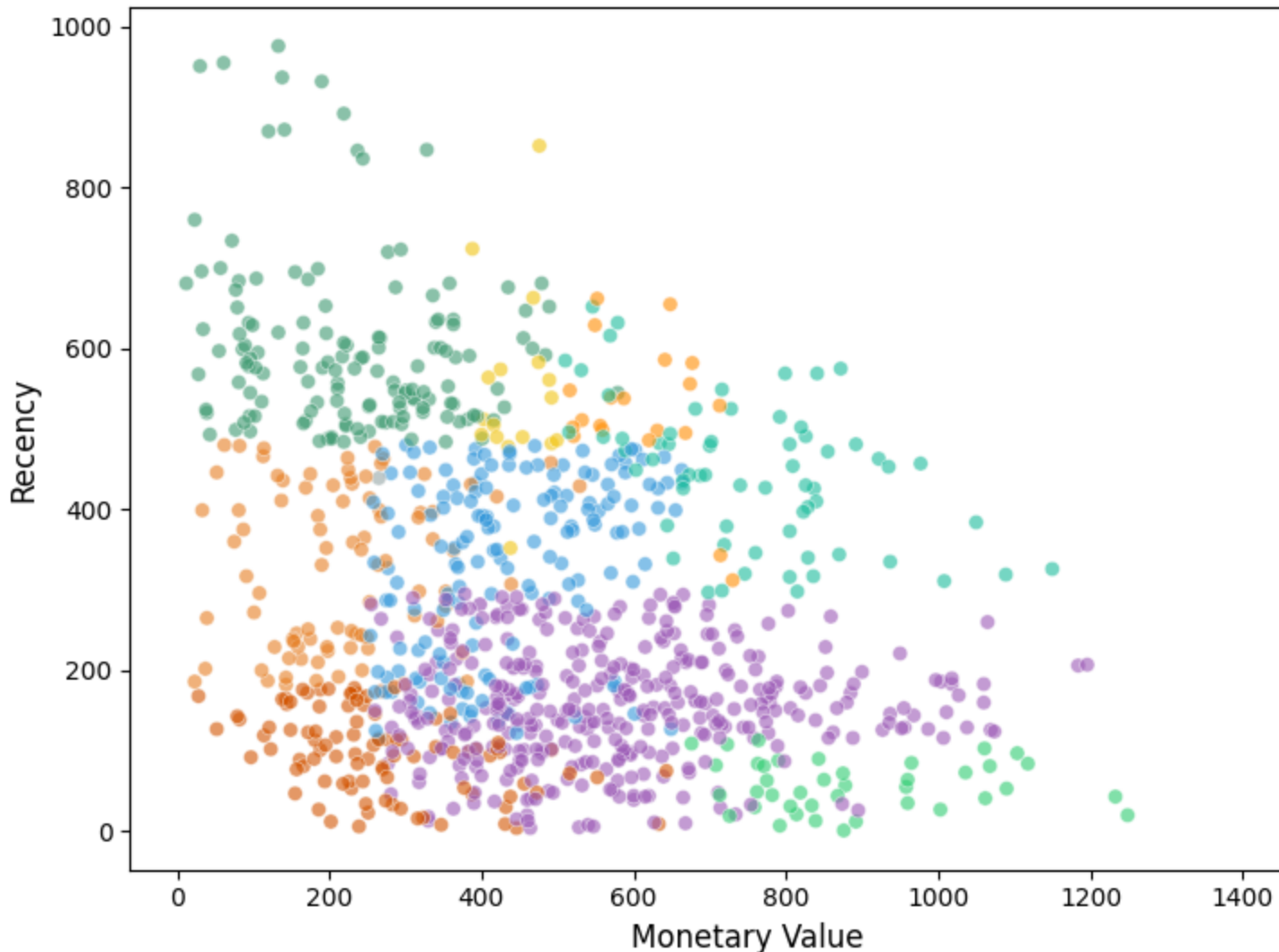


- **Most customers order 2–4 times:** Over 150 customers place 2, 3, or 4 orders, with 187 customers ordering 3 times.
- **Few customers order a lot:** Only a few customers place 10 or more orders, like 5 customers with 12 orders.
- **Some order just once:** 43 customers place only 1 order, which is a smaller group.
- **Orders drop after 5:** After 5 orders, the number of customers goes down, with only 20 customers ordering 8 times.
- **Focus on repeat buyers:** Most customers order a few times, so we should encourage them to order more often.

3.9 Relationship Between RFM Metrics

Recency vs. Monetary Value

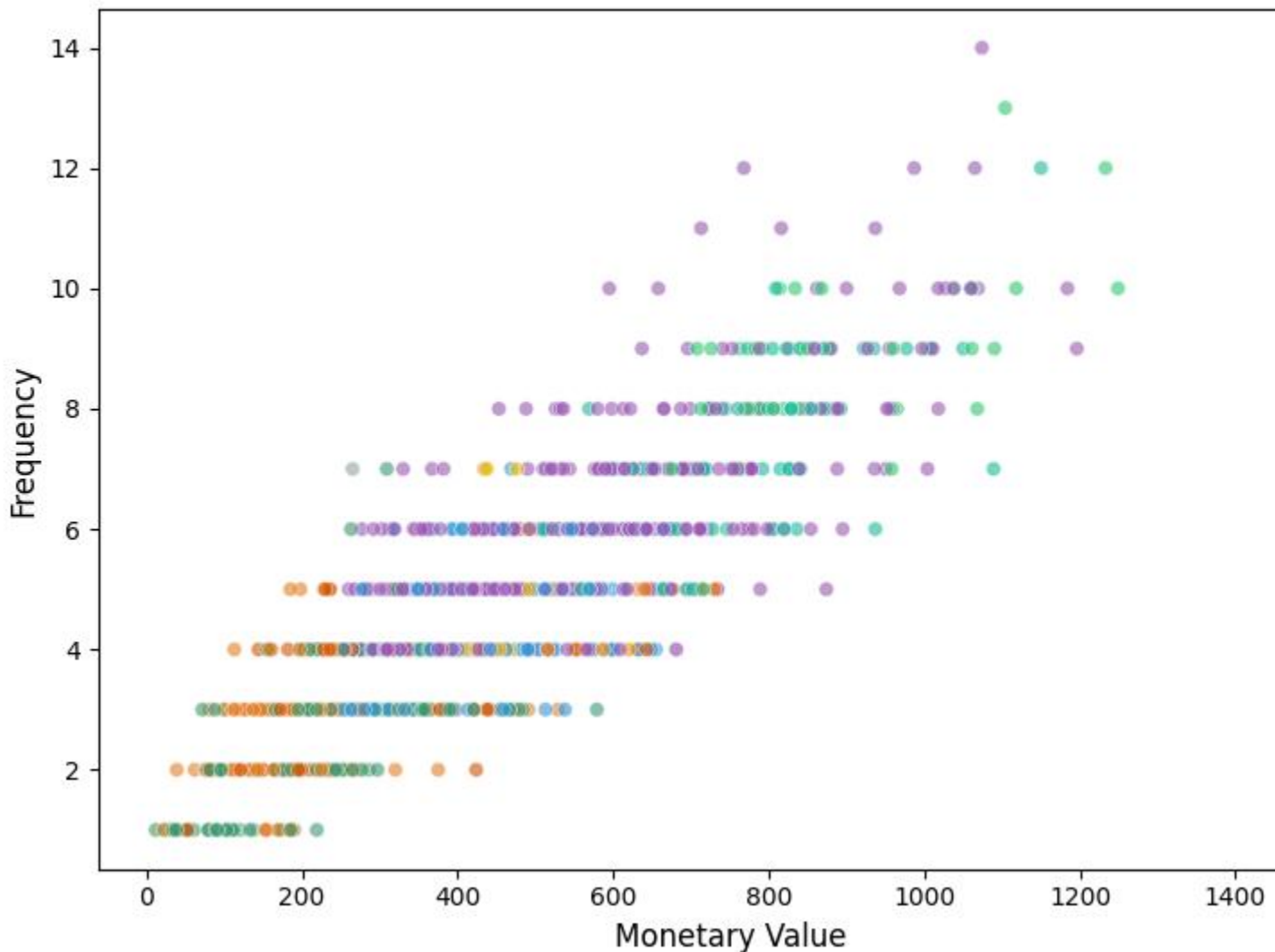
Recency vs Monetary Value



- **Many customers buy recently and spend less:** Most customers (like Potential Loyalists in purple) have low Recency (0–250 days) and spend \$0–\$600.
- **Some spend a lot but not recently:** Groups like Can't Lose Them (yellow) and Lost (green) spend \$500–\$2000 but haven't bought in a long time (500–1000 days).
- **Champions spend a lot and buy often:** Champions (green) have low Recency (0–250 days) and spend \$500–\$2000, making them the best customers.
- **New Customers spend little:** New Customers (blue) have low Recency (0–250 days) but spend less than \$500.
- **Lost customers need attention:** Lost (green) and About To Sleep (red) groups have high Recency (500+ days) and spend little, so we should try to bring them back.
- **Focus on keeping Champions:** Champions (green) are great because they buy often and spend a lot, so we should keep them happy.

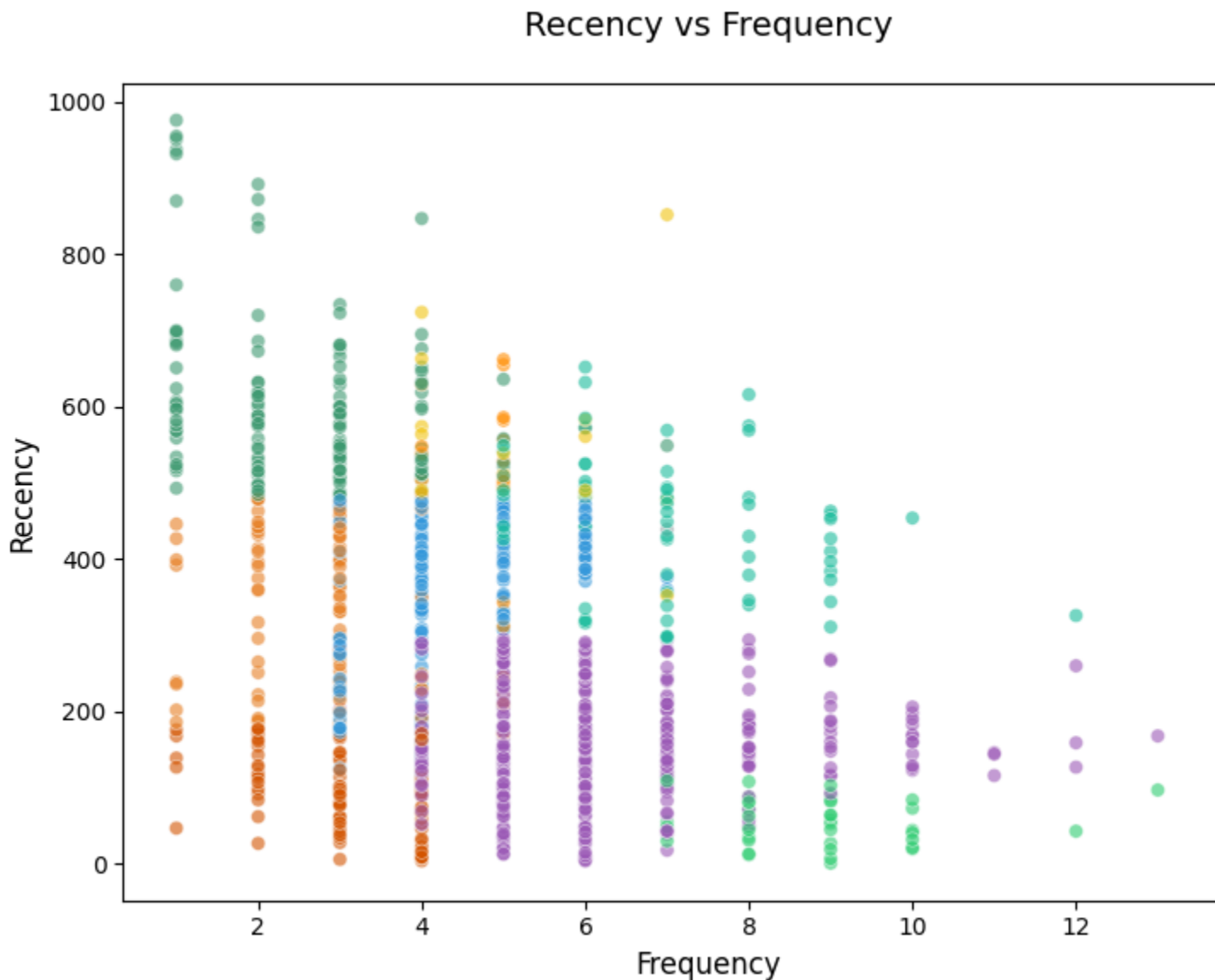
Frequency vs. Monetary Value

Frequency vs Monetary Value



- **Most customers buy 1–5 times:** Many customers, like Potential Loyalists (purple), buy 1–5 times and spend \$0–\$600.
- **Some buy a lot and spend a lot:** Champions (green) buy 8–14 times and spend \$600–\$1400, making them the best group.
- **Few buy often but spend little:** Some customers, like New Customers (blue), buy 1–5 times but spend less than \$400.
- **High spenders buy more often:** Customers spending \$1000+ (like Champions and Can't Lose Them) usually buy 8+ times.
- **Low frequency, low spenders:** Groups like About To Sleep (red) buy 1–3 times and spend less than \$400, so they need encouragement.
- **Focus on Champions:** Champions (green) buy often and spend a lot, so we should keep them happy to keep sales high.

Recency vs. Frequency



- **Most customers buy 1–5 times:** Many customers, like Potential Loyalists (purple), buy 4–14 times, with Recency from 0–300 days.
- **Some buy often but not recently:** Groups like Lost (green) and Can't Lose Them (yellow) buy 8–14 times but haven't bought in a long time (600–1000 days).
- **Champions buy often and recently:** Champions (green) buy 8–14 times and have low Recency (0–200 days), making them the best group.
- **New Customers buy less often:** New Customers (blue) buy 1–5 times and have low Recency (0–200 days).
- **Lost customers need attention:** Lost (green) and About To Sleep (red) groups have high Recency (400+ days) and buy less often, so we should try to bring them back.
- **Focus on Champions:** Champions (green) buy often and recently, so we should keep them happy to keep sales strong.

3.10 Segmentation Outcome

- **Strengths:** Potential Loyalists (36.48%) offer a chance to build loyalty due to their size and activity; Champions (4.02%) are ideal customers with high spending and frequency.
- **Challenges:** The large Lost segment (14.37%) and declining activity in At Risk and Can't Lose Them show a need to re-engage inactive customers.
- **Opportunity:** Low frequency and spending in groups like New Customers and About To Sleep suggest current strategies may not encourage enough repeat purchases or higher spending.

4. Strategy Recommendation

This section offers clear steps to improve customer engagement, keep valuable shoppers, and grow sales based on the RFM analysis of 4,906 transactions from 995 customers between 2004 and 2006. The ideas focus on how recently customers shopped (recency), how often (frequency), and how much they spent (monetary value).

- **Keep Top Shoppers Happy**
 - **Champions (40 customers, 4.02%):** These customers shop a lot (9 times), spend the most (\$846), and bought recently (54 days ago). Give them special treatment, like faster help or invites to exclusive events, to keep them coming back.
 - **Loyal Customers (160 customers, 16.08%):** They shop 4 times, spend \$413, and last bought 363 days ago. Offer them a small discount (like 10% off) to remind them to shop again and stay loyal.
- **Turn Good Shoppers into Great Ones**
 - **Potential Loyalists (363 customers, 36.48%):** This big group shops often (6 times), spends \$558, and last bought 152 days ago. Send them deals (like \$10 off their next buy) or early access to new items to make them shop more and stay with us.
 - **New Customers (96 customers, 9.65%):** They shop 3 times, spend \$241, and bought 101 days ago. Give them a welcome gift (like \$5 off their second buy) to get them excited and shopping more often.
- **Wake Up Slow Shoppers**
 - **About To Sleep (88 customers, 8.84%):** They shop 3 times, spend \$215, and haven't bought in 333 days. Send a "we miss you" note with a deal (like 10% off this week) to get them active again.
 - **Promising (3 customers, 0.30%):** This tiny group spends \$442 but last bought 203 days ago. Offer a simple coupon to keep them interested before they stop.
- **Bring Back Lost Shoppers**
 - **Lost (143 customers, 14.37%):** This big group hasn't shopped in 576 days and spends little (\$222). Send emails with big discounts (like 20% off) or free shipping, showing items they liked before, to win them back.

- **At Risk (20 customers, 2.01%) and Can't Lose Them (64 customers, 6.43%):** They haven't shopped in 520 and 451 days but spend \$603 and \$743. Send personal messages with quick deals (like "Buy in 5 days for 15% off") and ask why they stopped to bring them back.
- **Boost Shopping and Spending**
 - **More Shopping:** Most customers shop 2–5 times, so start a points system (like 10 points per \$1 spent, trade 500 points for \$10 off) to push groups like Potential Loyalists and New Customers to shop more.
 - **More Spending:** For low spenders like Lost (\$222) and About To Sleep (\$215), suggest extra items at checkout (like "Add this for \$5"). For big spenders like Champions, show fancier products to keep their spending high.

5. Conclusion

This study used the RFM model to analyze 4,906 transactions from 995 customers between 2004 and 2006. It aimed to group customers, understand their shopping habits, and suggest ways to improve marketing and relationships. The findings and steps show RFM helps us know our customers better and grow the business.

5.1 Key Findings

- **Exploratory Data Analysis (EDA) Results:** Before applying the RFM model, we conducted exploratory data analysis (EDA) to understand the dataset of 4,906 transactions from 995 customers between 2001 and 2006. The EDA showed that most transactions happened between 2004 and 2006, with only one customer in 2001, so we focused on the later years. We found 995 unique customers, with many shopping 2–5 times and spending less than \$200 on average. The data also had some gaps, like missing years, which we cleaned up to ensure accurate results.
- **Segmentation Criteria Comparison:** We tested two ways to group customers using the RFM model. The first method created 8 groups, but 128 customers (12.86%) ended up in an "Others" category, which was too large and unclear. The second method, which we chose, created 11 groups and reduced the "Others" category to just 1 customer (0.10%). This second method was better because it gave clearer groups that matched our data, making it easier to understand customer behavior.
- **Customer Groups:** Using the second method, we split 995 customers into 11 groups based on how recently they shopped, how often, and how much they spent.
- **Top Shoppers:** Champions (40 customers, 4.02%) are our best group—they shop 9 times, spend \$846, and bought 54 days ago, showing they're very active and valuable. Potential Loyalists (363 customers, 36.48%) shop 6 times, spend \$558, and bought 152 days ago, making them the largest group and a big chance to build loyalty.

- **Slow Shoppers:** Lost customers (143 customers, 14.37%) haven't shopped in 576 days and spend only \$222, while At Risk (20 customers, 2.01%) and Can't Lose Them (64 customers, 6.43%) stopped at 520 and 451 days ago, respectively. These groups show that many customers need help to start shopping again.
- **Shopping Trends:** Many customers shopped recently (within 100 days), which is a good sign, but some haven't bought in over 500 days, meaning activity dropped from 2004 to 2006. Most customers shop 2–5 times and spend under \$200, leaving room to encourage more frequent shopping and higher spending.

5.2 Recommendations and Future Steps

- **Action Plans:** The steps in Section 4 match these findings. For example, keep Champions happy with special perks, turn Potential Loyalists into loyal shoppers with deals, and wake up Lost customers with big discounts.
- **Grow Easy Wins:** Focus on New Customers and About To Sleep with small rewards to get them shopping more. Use points and extra items to boost shopping and spending for all groups.
- **Fix Slowdowns:** Check recency often to catch shoppers slowing down early and offer deals to keep them going, especially for groups like Lost and At Risk.
- **Goals Met:** This study grouped customers, showed their habits, checked trends from 2004 to 2006, and gave clear plans, hitting all its targets.
- **Next Moves:** Keep RFM data fresh by checking it every few months. Ask slow shoppers (like Lost) why they stopped to make our plans sharper and keep improving.

References

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